

Managing change within BT's operational support systems

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BT has around 1300 applications covering all aspects of its operations. The need to keep pace with an increasing rate of change in the telecommunications industry is being impacted by the ability to design and manage changes to such a large integrated applications portfolio. Computing technology and design methods are also advancing, and both old and new technologies have to work together. Achieving a balance between replacement and enhancement of applications within the constraints of money, time, and skills availability has to be carefully managed. New ways of looking at the portfolio have been developed so that a set of coherent strategies for the design and management of change can be implemented.

1. Introduction

BT's operational support systems (OSS) consist of the total systems capability that supports the sales, service, network and business operations of BT. They touch every aspect of BT's operations and are used, either directly or indirectly, by every member of the company. By the year 2000, the leaders in the telecommunications industry will include those companies who have successfully exploited their OSS to create and meet new market opportunities. Hence, leading-edge systems capability is critical to BT's long-term success.

BT's OSS capabilities are world class; the reliability and cost-effectiveness of the service, billing and network management applications are second to none. BT's customer service system (CSS) remains one of the most highly integrated customer service applications deployed by any PTT and has been successfully adapted to serve new products and support several new BT organisational structures. The deployment in 1995 of elapsed time charging within the billing architecture provides the BT group with a highly flexible infrastructure with which to face a rapidly changing competitive scene. The operations management centre (OMC) system provides comprehensive switch management capabilities, and its links to CSS provide automated service configurations for the public switched telephone network (PSTN).

The OSS portfolio has been developed over the last 20 years, growing from applications which were sponsored by functional and organisational groups. Today those systems have been progressively interlinked in order to meet ever-rising service standards. This integration has in itself brought its own issues that need to be addressed. The family

of systems contains too much diversity of data and infrastructure. Additionally their original organisation, product and customer segment focus have in many cases been affected by rapid change in those dimensions. System development capabilities are struggling to keep pace with the commercial drivers and in parts of the company are perceived to be a barrier to the fast introduction of new products and services, delivery of service excellence, or support of global ambitions. The data is locked into those systems preventing exploitation of the information needed to maintain and retain BT's market share.

In numerical terms the systems portfolio over the last five years has doubled the number of applications to around 1300, and tripled the number of interfaces between applications to around 3000, with each interface typically being hand-crafted to meet a specific business need (see Fig 1). It is estimated that the portfolio contains in excess of 80 million lines of code with an annual increase of 8 million new lines of code written and 4 million lines of code replaced.

A recent review has identified about 30 major systems that are critical to the operations of the company and account for the majority of system operating costs. These are called hub systems (see Fig 2). Surrounding these hubs are a much larger set of systems manipulating and feeding data into these hubs. The remaining systems tend to be stand-alone, with manual, rather than automated, links to other systems. This categorisation of systems has started the debate on where the limited amount of development

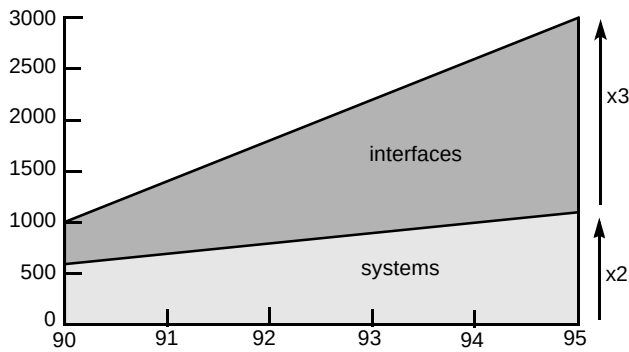


Fig 1 System and interface growth.

resource should be allocated, and where the priorities for re-engineering lie.

The remainder of this paper reviews the drivers causing change to systems, the management and technical vision for systems, and the approach being taken to ensure that systems continue to meet the ever-increasing demands.

2. Drivers on OSS

The telecommunications industry is rapidly changing with the emphasis very much on reducing the time to market, lower whole-life costs, and a shift of focus towards customers and service, while moving away from network technology and technology-related identifiers, e.g. telephone numbers. Systems must therefore be capable of supporting change without having to go through a lengthy design, develop, test and release life cycle, and be deployed on standard rather than bespoke infrastructure. This must be achieved at a price that is appropriate to the product and service being delivered or the internal process being supported.

Regulation is having an impact upon OSS either as new requirements, such as number portability (which requires

specific telephone numbers to be treated as a resource that can be passed between licensed operators), or increasing the interaction between BT and other operators with equal access and OLO interconnect. System designs for the future must therefore ensure that systems can be linked to other systems outside BT while protecting commercial interests and meeting the legal obligations of protecting data relating to customers or their installations.

Partnerships, joint ventures and strategic alliances are being formed to extend or open up markets. These are often based on the linking of computer systems, sharing systems between partners, or using common software as a means to reduce or stabilise costs, share risk, and improve time to market. The OSS of the future will need to cater for all of these needs and will require clear business decision making to provide direction and priorities.

Network technology changes, with the introduction of higher bandwidth and intelligent components into the network, will enable new products and services to be created and configured much more rapidly than at present. These new technologies, together with the services they promise, will require changes to the present OSS base if their features are to be exploited and mass customisation (i.e. services packaged and tailored to an individual's needs) is to be delivered.

Information technology is changing rapidly with year-on-year price/performance improvements and increasing commoditisation. A mixed technology infrastructure where different suppliers' products, both hardware and software, have to coexist and work together is rapidly becoming the norm. Personal computers are being increasingly deployed as 'front-ends' to existing systems to provide additional intelligence, work-flow features, data and linkages between

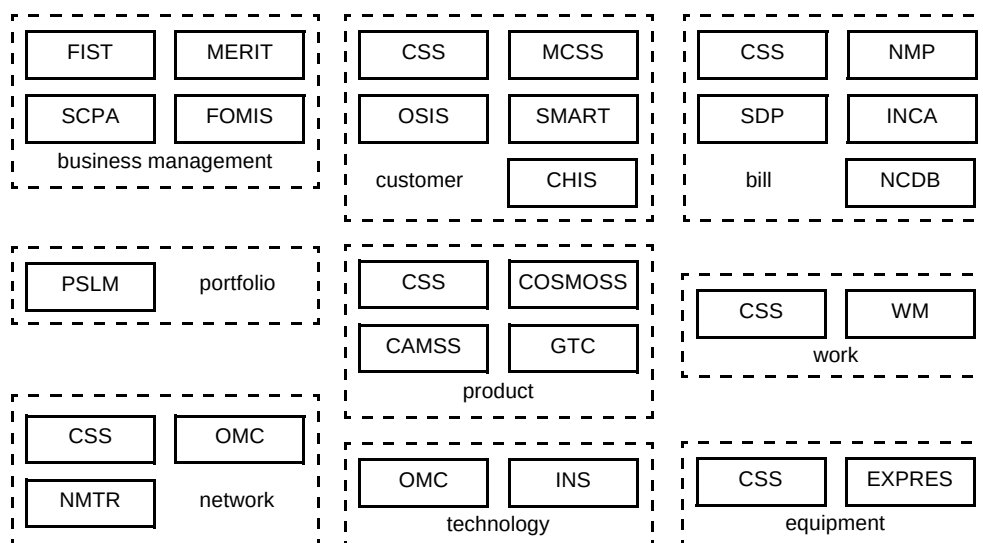


Fig 2 Hub systems.

systems. However, these gains from advances in technology may be offset by the increasing demand on computing operations management in a distributed-processing and data-storage environment. The Internet, particularly with the ability to write, distribute and run applications using tools like Java™, has come of age as a commercially viable infrastructure that will deliver service reliably. The question is how to make best use of it.

3. OSS management vision

In management terms the OSS portfolio is a key enabler for the success of BT. The vision is that OSS will not be a barrier to change, will never appear on the critical path for new services, and will provide flexible and consistent solutions.

The goals are for changes to OSS to occur in less than:

- 30 days for minor changes to existing products and services,
- 90 days for major changes to existing products and services,
- 120 days to support new services.

The achievement of this vision, which is extremely challenging when looking, for example, at the best time-scales for delivery of changes today, is dependent upon the ability to design and deploy a systems and management structure that is capable of controlling change happening at an order of magnitude greater than at present.

4. Technical vision

The need to partition BT's systems into small modules to provide the flexibility, delivery -time improvements and exploitation of computing technology advances requires a different architecture to the mainframe and limited intelligence terminal that have served BT well over the last two to three decades (see Fig 3). Industry thinking has been moving towards a three-tier model that separates out the functionality associated with the terminal, from that associated with the 'true' processing and also from that functionality required to manipulate and protect the integrity of data

While methods have been available for many years to design systems along these lines it is only over that last year or so that the operating system software capable of controlling such an environment has matured enough and been reliable enough to be used in large-scale production environments. Even having said that, the support tools to maintain and control an environment containing distributed processing and distributed data are still in their infancy and may well, together with the new skills and techniques needed, be the constraining factors in the implementation of this architecture.

The aim, however, is to build a three-tier client/server architecture using a standard IT infrastructure built from a small set of proven off-the-shelf components. The tiers will, in general, align to the model of presentation, function, and data [1].

To capitalise upon existing assets, data will be held and managed in 'back-end' systems developed by the evolution of existing hub systems. Similar or duplicate data that is

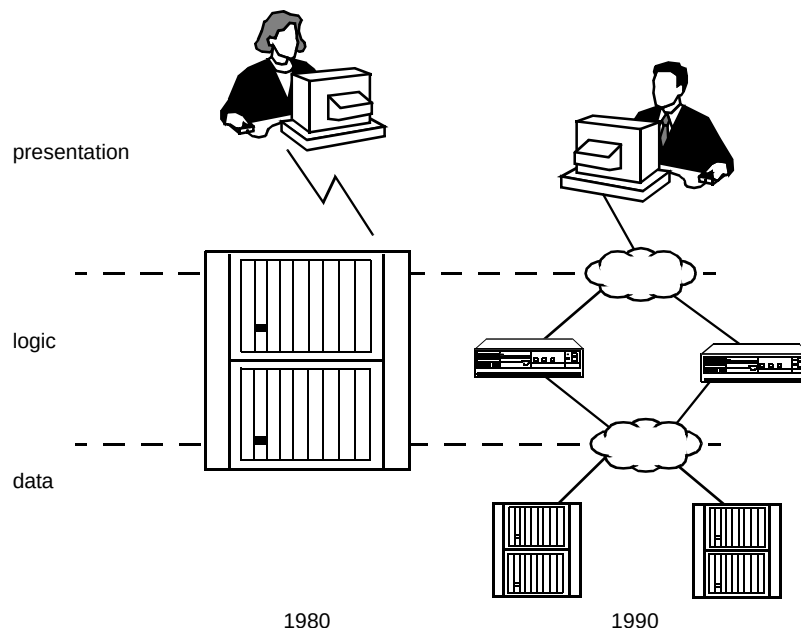


Fig 3 Comparison of old and new architectures.

currently locked into different systems will be released either by developing a specific database designed to meet future needs, or made accessible by implementing data gateways that direct applications to a database. A current example of this type of development is the operational customer database project which will, over a period of time, bring together customer data currently dispersed in systems like CSS [2] and COSMOSS [3]. The main business benefits will be that this data is now available to many applications (and not just the one associated with the database), alternative sources of the same data can be eliminated, the data can be cleansed, and, as there is one source, the maintenance of the data is simplified. Where a single database is not achievable, for example where delays in accessing the data would be unacceptable, duplicates could be created with automated processes to maintain a master/slave relationship to preserve data integrity.

The middle tier will consist of a set of self-contained 'services' or 'objects' that can be called upon to perform a specific function, e.g. 'Get me account details for this customer', or 'Configure this line for these network services'. Each service will be designed to fulfil a specific need identified by an analysis of current functions and future requirements. They would be independent of other services and capable of being changed without affecting other services or function in the other tiers. They would have published interfaces into and out of the service by which they communicate to other tiers or other services (see Fig 4). They will be defined so that they can be reused to support many products and services and be capable of being assembled to support as yet unknown needs. Development of these mid-tier services will either be bespoke or by re-engineering them out of their existing back-end system. Each service will have a range of deployment options and can then be replicated and distributed over the standard IT infrastructure as demand for that service changes.

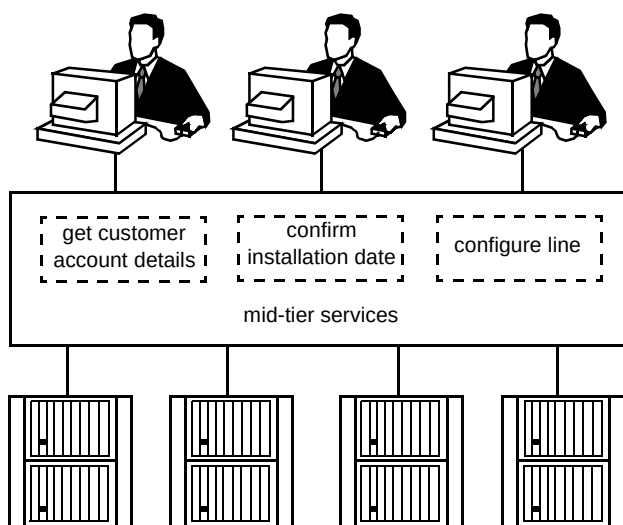


Fig 4 Mid-tier services linked to presentation and data tiers.

Mid-tier services are self-contained groupings of functions and data that can be called using known interfaces (see Fig 5).

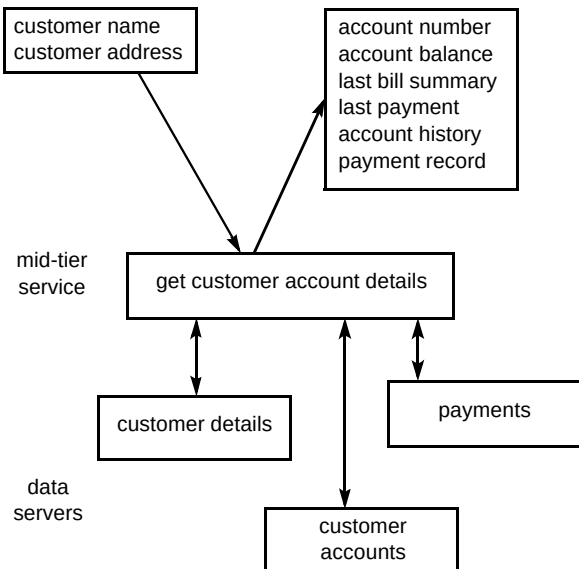


Fig 5 Example of mid-tier service interfaces.

The front-end 'clients' will contain applications that present the services, say, on a terminal screen, and manage the order in which the services are offered. In their current form these clients exist as personal computers or telephone keypads. In the future they will exist in many other forms and be capable of being operated, if this is what they want, directly by customers to initiate an order or change of existing services directly into BT's OSS. The aim is for this to happen with no or little involvement from BT people. For example, a customer may initiate the call-off of additional bandwidth from a terminal on their premises directly into the OSS, the service configured, records updated, and billing triggered, without any BT person being involved.

5. Approach taken

The tile architecture defined in the BT OSS architecture [4] sets the overall direction for systems planning and ensures close links between the evolution of systems and the visionary processes that this company could operate. The tiles provide a map showing how functions to support all the operational needs of BT can be grouped around specific sets of data. Mapping today's systems on to these tiles clearly brings into focus many of the difficulties faced when trying to integrate systems to meet today's needs. Using the tiles as a template, a series of top-level designs, known as tile evolution plans, have been defined for each tile. These plans have provided systems designers with a

high-level evolution path spanning four to five years. Designs that follow these evolution plans will ultimately lead to a much simplified systems structure built from a rationalised OSS portfolio.

For each hub system a review of current and planned capabilities was undertaken by their chief designer. This exercise brought out a business view of what products and services were being or were planned to be supported, as well as the data used, technology employed, and existing or future interfaces. These systems evolution plans forced a longer term view of systems, out beyond the next two or three releases, and enabled key changes to be identified that would start the re-engineering of these systems towards the three-tier architecture. Additionally, disjoints between system plans were made visible much earlier than they would have been if the requirements specification route had been followed. This enabled corrective action to be taken before the development processes started.

The next step has been to consider the design of future OSS in business rather than system terms. Broad-level 'business activities' such as customer handling, billing, provisioning and work management have been defined. By gathering the management of systems within these domains, both design and development can be co-ordinated to deliver, over a period of time, a set of integrated systems. This grouping also provides a forum in which business needs and priorities can be discussed and agreed. Decisions, such as which system will be enhanced to deliver new functionality, can then be based on pan-organisational needs and implemented in a way that still enables progress to be made towards the future architecture.

Using this approach, the major (hub) systems in the OSS portfolio will evolve to become the 'back-end' data management engines. Those functions that manipulate the data, read or write data to the database, or carry out data integrity checking will remain. Other processing functions will be re-engineered, or replaced by new code, into services that will be deployed in the mid-tier. This has already started with the customer-handling interface server (CHIS) project which has delivered a set of encapsulated services, originally CSS transactions, to provide simple network services. Interestingly the client tier for this application is a telephone with voice commands from a speech application processor. This application has been deployed on a retailer's premises and requests for changes to certain of BT's select services can now be carried out by the retailer directly, using the telephone, a speech application processor [5], the re-engineered services, CSS and OMC [6] directly into the network switch [1].

This strategy will reduce the number of systems by directing new functions, and by re-directing existing

functions and data (if still required), currently realised in non-hub systems, towards hub systems. The evolution plans will form the basis for tracking the delivery of new capabilities and progress towards system simplification.

Collaboration with partners will be actively pursued to reduce costs. This will take many forms from using a partner's application, to sharing the development risk through a joint development. The OSS of the future will therefore contain a mixture of commercial packages, partners' applications, joint development and (internally developed) bespoke applications. Harmonious working to deliver 'seamless' service will require conformance to inter-working, data and technical standards that has never been achieved to date.

Forming strategic partnerships with suppliers will provide long-term benefits for both parties — for BT by having access to a supplier's skills and expertise and the ability to focus the supplier on BT's development needs, and for the supplier by having an outlet for their products and an insight into the future direction and needs of BT.

Technical standards and design practice that will be used are being defined. A sub-set of technologies, covering hardware, operating systems, network service/protocols, system management, enabling components (distribution services, data management, presentation services, etc) and development tools, has been selected, from which future systems will be constructed. An implementation plan is in place to educate designers and engineers about this standard portfolio (see Fig 6). The move towards this set of technical standards will take place over many years, as, for example, it is unlikely that a database would be re-engineered to conform to a standard database management system, if it was satisfactorily performing and still being supported by the supplier.

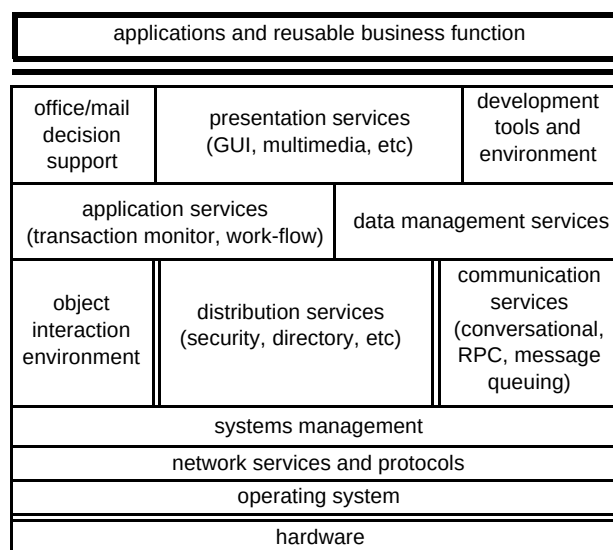


Fig 6 Standard IT infrastructure framework.

6. Conclusions

The demands being placed on BT's OSS are escalating. Satisfying these demands requires step changes over and above those achieved by implementing improvements to what already exists. The new ways of looking at the structure of OSS developed over the last two years provide the framework to ensure that what is designed and developed meets business needs and will form an integrated OSS portfolio. Re-engineering existing hub systems either as back-end data servers or mid-tier services will make best use of existing assets. Implementation of a three-tier architecture is the radical change that will deliver the flexibility and radical reductions in the development life-cycle times. Achieving these key business goals will depend on a technically and managerially challenging design and development discipline.

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